

Ankur Desai

(248) 657 – 3805 | ardusa05@gmail.com | www.ardusa.dev | linkedin.com/in/ardusa

EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

MindMosaic

Software Engineering Intern

Aug. 2025 – May 2026

East Lansing, MI

- Built and shipped an interactive knowledge-graph platform that keeps dense, interconnected content navigable as it grows, delivering a production MVP to real users on a polyglot Spring Boot, Neo4j, and PostgreSQL stack.
- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Authored an AWS deployment script and managed the environment as code with Terraform, turning a manual multi-service bring-up into a repeatable, version-controlled provision so the polyglot stack deploys identically every time.

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Built the full language-to-motion pipeline in simulation on ROS2 and Gazebo with the attack layer decoupled at the ROS2 interface, so swapping the target robot platform cost only the URDF and control layer and never touched the security result.
- Built the defense under attack as a steelman, composing a triage-plus-verification gate from three published safety architectures rather than a naive single classifier, so the result survives the objection that a real system would do more.

Robostangs (FRC Team 548)

Software Captain

Jun. 2023 – May 2024

Northville, MI

- Led the software and electrical teams (5 and 6 engineers) for a competition robot that reached the FIRST World Championship and placed top 16 in its division in Houston, the program's strongest world-stage finish since 2014.
- Anchored auto-aim on real-time, vision-corrected localization that fused two AprilTag cameras with swerve odometry on a dedicated 250Hz thread with latency compensation, holding pose accurate during aggressive motion.

PROJECTS

Dadei | *Python, FastAPI, Redis, PostgreSQL, Docker*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Routed every model-proposed side effect through a single approval chokepoint, so the assistant never fires an irreversible calendar, email, or task action without a cancelable countdown, giving every real-world effect one auditable seam between the LLM and the outside world.
- Decoupled background workers from the live API behind a Redis Streams event bus with consumer groups, dead-letter routing after five failed deliveries, and stale-claim recovery, giving at-least-once delivery across process boundaries that Redis pub/sub silently dropped on disconnect.

fliks | *Go, PostgreSQL, AWS EC2*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Coordinated transcode jobs through a Postgres-backed queue using SELECT FOR UPDATE SKIP LOCKED instead of bolting on Redis or SQS, keeping the whole system on one datastore and one self-provisioned EC2 box with no managed queue or transcoding service.

TECHNICAL SKILLS

Languages Python, Java, Go, C++, JavaScript/TypeScript, SQL, Bash
Cloud & Infrastructure AWS (EC2, S3, RDS), Docker, Terraform
Frameworks Spring Boot, FastAPI
Databases PostgreSQL, Neo4j, Redis
Robotics ROS2, Gazebo, AprilTag